Topic quiz: Work and energy

IBDP Physics SL · Grade 11 · A.3 Work and energy · 10 multiple-choice questions

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Name: Class: Date:

How to use. Practise with the simulations and worksheets first. Then answer every question: circle one letter and write it in the grid below. Use g = 9.81 m s^{-2} and show any working next to the question.
Check yourself online: the same quiz with instant feedback and explanations — https://school.taneski.com/physics/dp/quiz-work-energy/

1	2	3	4	5	6	7	8	9	10

Question 1

A child pulls a sled 10.0 m along flat snow with a force of 50.0 N at 30° above the horizontal. How much work does the pulling force do?

(A) 43.3 J
(B) 250 J
(C) 433 J
(D) 500 J

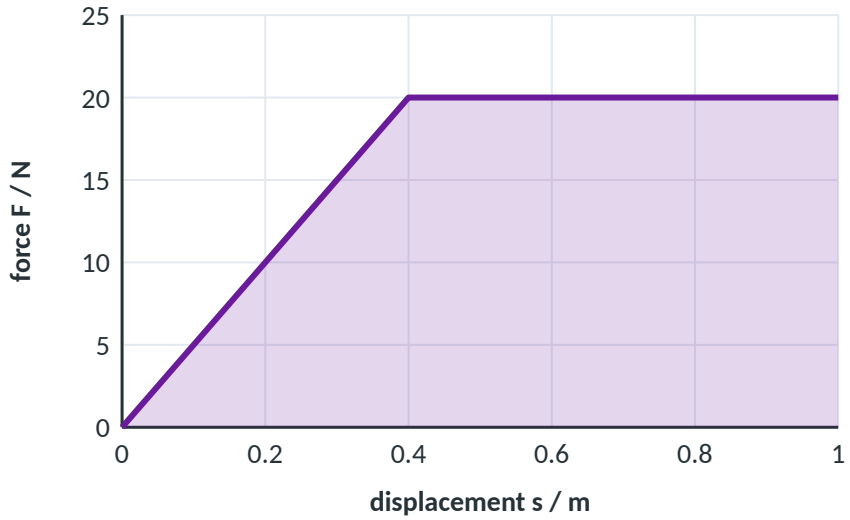
Question 2

A box is pulled 5.0 m along a horizontal floor. How much work does the normal force do on the box?

(A) Zero, because the normal force is perpendicular to the displacement
(B) Zero, because the normal force is cancelled by the weight
(C) mg × 5.0 m
(D) A negative amount, because it opposes the motion

Question 3

The graph shows the force on a trolley against its displacement. How much work is done over the first 1.0 m?



(A) 4 J
(B) 12 J
(C) 16 J
(D) 20 J

Question 4

A 20 kg box is raised 1.5 m, either by lifting it straight up or by pushing it up a 3.0 m frictionless ramp at constant speed. Which statement is correct?

(A) The ramp needs twice the work, because the distance is twice as long
(B) The ramp needs half the work, because the force is half as big
(C) Lifting needs no work, because the box ends at rest
(D) Both need the same work, 294 J, but the ramp needs a smaller force

Question 5

A car brakes from 20 m s^{-1} and stops in 25 m. With the same braking force, what is its stopping distance from 40 m s^{-1}?

(A) 25 m
(B) 50 m
(C) 100 m
(D) 200 m

Question 6

A ball is dropped from a height of 5.0 m. Air resistance is negligible. How fast is it moving just before it hits the ground?

(A) 7.0 m s^{-1}
(B) 9.9 m s^{-1}
(C) 49 m s^{-1}
(D) 98 m s^{-1}

Question 7

A 0.500 kg Smart Cart rolls down a ramp. It loses 0.513 J of gravitational potential energy and gains 0.483 J of kinetic energy. How much energy is transferred to thermal energy by friction?

(A) 0 J, because energy is conserved
(B) 0.030 J
(C) 0.483 J
(D) 0.996 J

Question 8

A heavy pendulum is released from rest touching a student's nose. The student keeps perfectly still. Why can the pendulum not hit the student when it swings back?

(A) It cannot rise higher than where it started, because its total energy cannot increase
(B) It loses all its energy at the far end of the swing
(C) Its mass is too large for it to come back
(D) Gravity is weaker on the way back

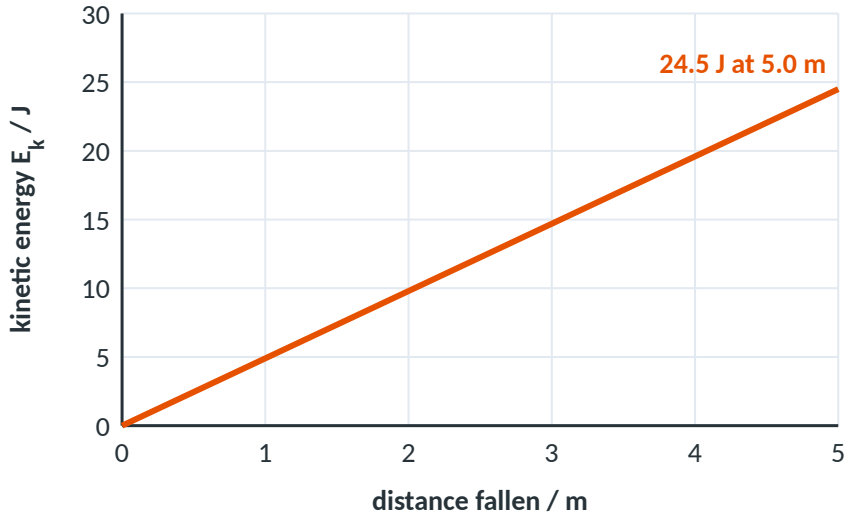
Question 9

A spring ($k = 200\ N\ m^{-1}$) is compressed by 0.050 m and launches a 0.020 kg toy straight up. Ignoring losses, how high does the toy rise?

(A) 0.64 m
(B) 1.27 m
(C) 2.55 m
(D) 51.0 m

Question 10

A 0.50 kg ball is dropped. The graph shows its kinetic energy against the distance fallen. What does the gradient of the graph represent?



(A) The ball's speed, 4.9 $m\ s^{-1}$
(B) The ball's mass, 0.50 kg
(C) The acceleration of free fall, 9.81 $m\ s^{-2}$
(D) The ball's weight, 4.9 N

End of quiz · Score: _____ / 10